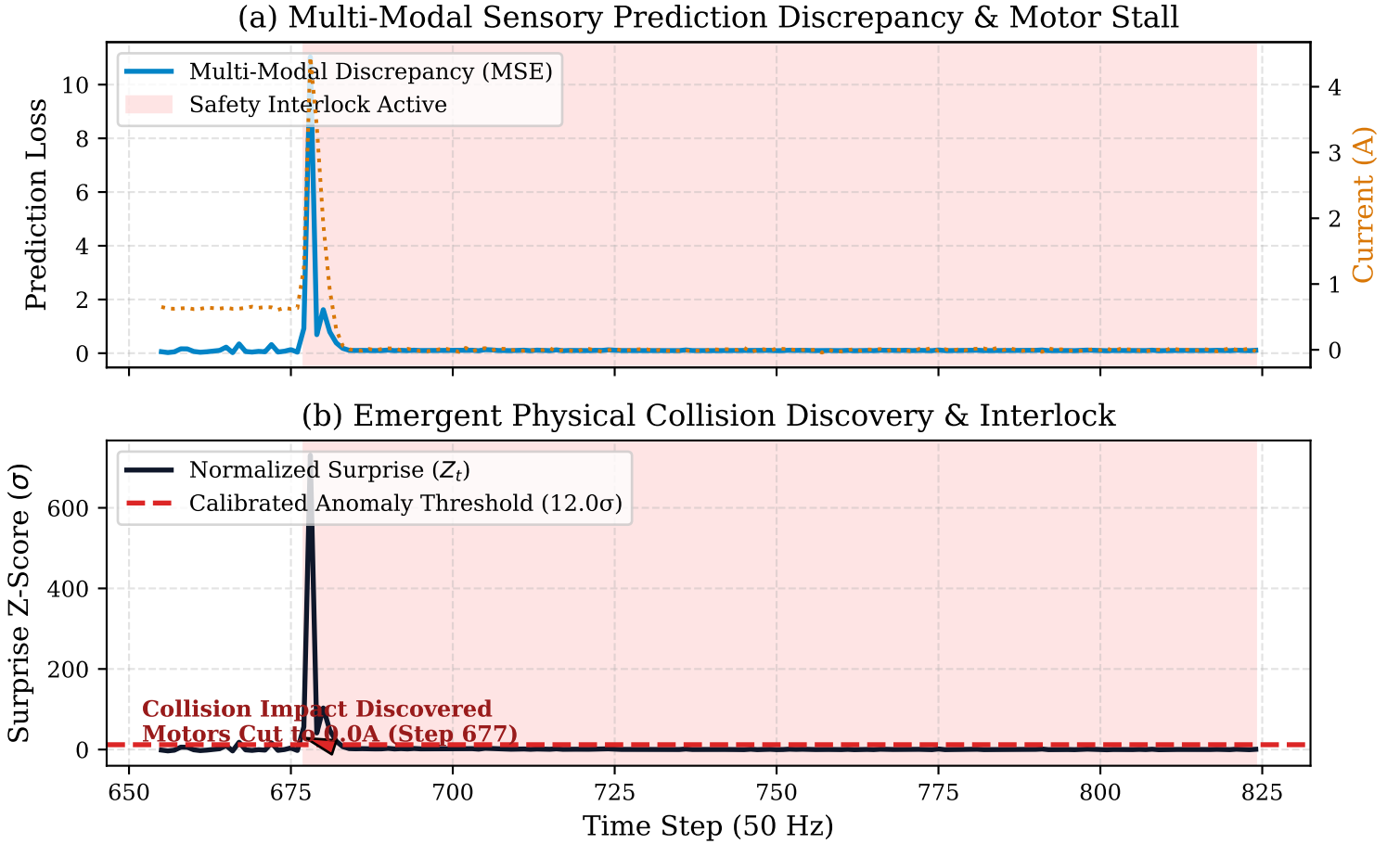
(a) Multi-Modal Sensory Prediction Discrepancy & Motor Stall
Multi-Modal Discrepancy (MSE)
Safety Interlock Active
Prediction Loss
Current (A)
0
2
4
6
8
10
0
1
2
3
4
(b) Emergent Physical Collision Discovery & Interlock
Normalized Surprise (Z_t)
Calibrated Anomaly Threshold (12.0σ)
Surprise Z-Score (σ)
Collision Impact Discovered
Motors Cut to 0.0A (Step 677)
0
200
400
600
650
675
700
725
750
775
800
825
Time Step (50 Hz)